CSE370 : Database Systems

Assignment 02 | Fall 2025

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No 1 Query (as Plain Text)

select first_name, last_name, email, phone_number, hire_date, department_id
from Employee where hire_date = (select max(hire_date) from Employee);

No 1 SS (of Query & Output in Shell)

```
MariaDB [Company_24141252]> select * from Employee;
+-------------+------------+-----------+-------------------------+--------------+-----------+--------+--------+-----------------+-----------+---------------+
| employee_id | first_name | last_name | email                   | phone_number | hire_date | job_id | salary | commission_pct | manager_id | department_id |
+-------------+------------+-----------+-------------------------+--------------+-----------+--------+--------+-----------------+-----------+---------------+
| EMP001      | John       | Smith     | john.smith@company.com  | 01712345001  | 2020-01-15 | JOB001 |  45000 |           0.150 | MNG001    | DPT001        |
| EMP002      | Sarah      | Johnson   | sarah.j@company.com     | 01712345002  | 2020-03-20 | JOB002 |  52000 |           0.200 | MNG001    | DPT001        |
| EMP003      | Michael    | Brown     | michael.b@company.com   | 01712345003  | 2021-05-10 | JOB003 |  28000 |           0.100 | MNG002    | DPT002        |
| EMP004      | Emma       | Davis     | emma.d@company.com      | 01712345004  | 2021-06-15 | JOB001 |  26000 |           0.120 | MNG002    | DPT002        |
| EMP005      | James      | Wilson    | james.w@company.com     | 01712345005  | 2019-08-22 | JOB004 |  38000 |           0.180 | MNG003    | DPT003        |
| EMP006      | Olivia     | Martinez  | olivia.m@company.com    | 01712345006  | 2022-02-18 | JOB002 |  48000 |           0.250 | MNG003    | DPT003        |
| EMP007      | William    | Garcia    | william.g@company.com   | 01712345007  | 2022-11-30 | JOB005 |  32000 |           0.140 | MNG004    | DPT005        |
| EMP008      | Sophia     | Rodriguez | sophia.r@company.com    | 01712345008  | 2022-11-30 | JOB006 |  29000 |           0.160 | MNG004    | DPT005        |
| EMP009      | David      | Lee       | david.lee@company.com   | 01712345009  | 2023-01-12 | JOB007 |  25000 |           0.080 | MNG005    | DPT007        |
| EMP010      | Isabella   | White     | isabella.w@company.com  | 01712345010  | 2023-01-12 | JOB008 |  27000 |           0.090 | MNG005    | DPT007        |
| EMP011      | Daniel     | Harris    | daniel.h@company.com    | 01712345011  | 2020-09-05 | JOB001 |  42000 |           0.170 | MNG001    | DPT001        |
| EMP012      | Mia        | Clark     | mia.c@company.com       | 01712345012  | 2021-11-20 | JOB003 |  29500 |           0.110 | MNG002    | DPT002        |
| EMP013      | Lucas      | Lewis     | lucas.l@company.com     | 01712345013  | 2022-04-08 | JOB004 |  36000 |           0.190 | MNG003    | DPT003        |
| EMP014      | Ava        | Walker    | ava.w@company.com       | 01712345014  | 2023-01-12 | JOB009 |  31000 |           0.075 | MNG005    | DPT007        |
| EMP015      | Ethan      | Hall      | ethan.h@company.com     | 01712345015  | 2022-07-25 | JOB010 |  29000 |           0.170 | MNG004    | DPT005        |
+-------------+------------+-----------+-------------------------+--------------+-----------+--------+--------+-----------------+-----------+---------------+
15 rows in set (0.001 sec)

MariaDB [Company_24141252]> select first_name, last_name, email, phone_number, hire_date, department_id
    -> from Employee where hire_date = (select max(hire_date) from Employee);
+------------+-----------+-----------------------+--------------+-----------+---------------+
| first_name | last_name | email                 | phone_number | hire_date | department_id |
+------------+-----------+-----------------------+--------------+-----------+---------------+
| David      | Lee       | david.lee@company.com | 01712345009  | 2023-01-12 | DPT007        |
| Isabella   | White     | isabella.w@company.com | 01712345010  | 2023-01-12 | DPT007        |
| Ava        | Walker    | ava.w@company.com     | 01712345014  | 2023-01-12 | DPT007        |
+------------+-----------+-----------------------+--------------+-----------+---------------+
3 rows in set (0.006 sec)
```

No 2 Query (as Plain Text)

select first_name, last_name, employee_id, phone_number, salary, department_id
from Employee e1 where salary = (select min(salary) from Employee e2
where e2.department_id = e1.department_id);

No 2 SS (of Query & Output in Shell)

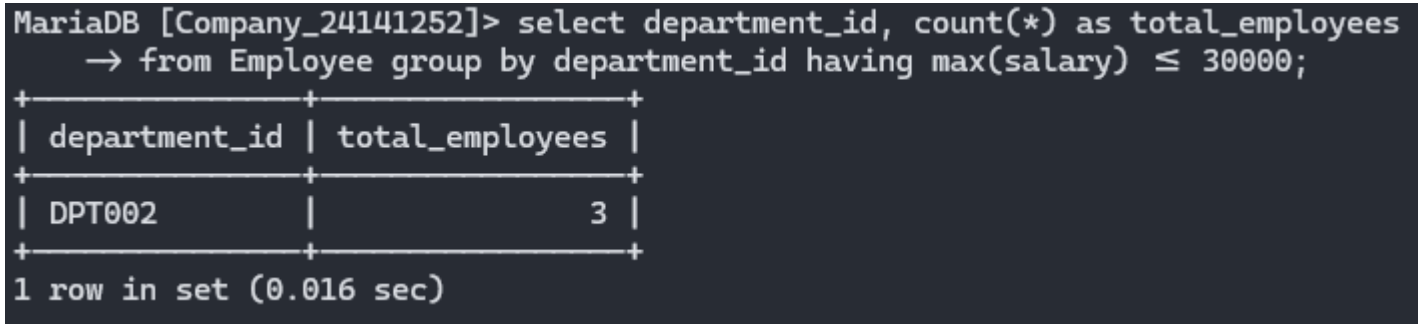
```
MariaDB [Company_24141252]> select first_name, last_name, employee_id, phone_number, salary, department_id
    -> from Employee e1 where salary = (select min(salary) from Employee e2
    -> where e2.department_id = e1.department_id);
+------------+-----------+-------------+---------------+--------+---------------+
| first_name | last_name | employee_id | phone_number  | salary | department_id |
+------------+-----------+-------------+---------------+--------+---------------+
| Emma       | Davis     | EMP004      | 01712345004   |  26000 | DPT002        |
| Sophia     | Rodriguez | EMP008      | 01712345008   |  29000 | DPT005        |
| David      | Lee       | EMP009      | 01712345009   |  25000 | DPT007        |
| Daniel     | Harris    | EMP011      | 01712345011   |  42000 | DPT001        |
| Lucas      | Lewis     | EMP013      | 01712345013   |  36000 | DPT003        |
| Ethan      | Hall      | EMP015      | 01712345015   |  29000 | DPT005        |
+------------+-----------+-------------+---------------+--------+---------------+
6 rows in set (0.001 sec)
```

No 3 Query (as Plain Text)

select first_name, last_name, employee_id, commission_pct, department_id
from Employee where department_id = 'DPT007'and commission_pct < all (select commission_pct
from Employee where department_id = 'DPT005');

No 3 SS (of Query & Output in Shell)

```
MariaDB [Company_24141252]> select first_name, last_name, employee_id, commission_pct, department_id
    -> from Employee where department_id = 'DPT007'and commission_pct < all (select commission_pct
    -> from Employee where department_id = 'DPT005');
+------------+-----------+-------------+----------------+---------------+
| first_name | last_name | employee_id | commission_pct | department_id |
+------------+-----------+-------------+----------------+---------------+
| David      | Lee       | EMP009      |          0.080 | DPT007        |
| Isabella   | White     | EMP010      |          0.090 | DPT007        |
| Ava        | Walker    | EMP014      |          0.075 | DPT007        |
+------------+-----------+-------------+----------------+---------------+
3 rows in set (0.017 sec)
```

No 4 Query (as Plain Text)	select department_id, count(*) as total_employees from Employee group by department_id having max(salary) <= 30000;
No 4 SS (of Query & Output in Shell)	 MariaDB [Company_24141252]> select department_id, count(*) as total_employees → from Employee group by department_id having max(salary) ≤ 30000; +---+---+ | department_id | total_employees | +---+---+ | DPT002 | 3 | +---+---+ 1 row in set (0.016 sec)
No 5 Query (as Plain Text)	select distinct e1.department_id, e1.job_id, e1.commission_pct from Employee e1 where e1.commission_pct < any (select e2.commission_pct from Employee e2 where e2.department_id = e1.department_id and e2.job_id != e1.job_id);

No 5 SS
(of Query & Output in Shell)

```
MariaDB [company_24141252]> select distinct e1.department_id, e1.job_id, e1.commission_pct from Employee e1
    -> where e1.commission_pct < any (select e2.commission_pct from Employee e2
    -> where e2.department_id = e1.department_id and e2.job_id ≠ e1.job_id);
+---------------+--------+----------------+
| department_id | job_id | commission_pct |
+---------------+--------+----------------+
| DPT001        | JOB001 |          0.150 |
| DPT002        | JOB003 |          0.100 |
| DPT003        | JOB004 |          0.180 |
| DPT005        | JOB005 |          0.140 |
| DPT005        | JOB006 |          0.160 |
| DPT007        | JOB007 |          0.080 |
| DPT001        | JOB001 |          0.170 |
| DPT002        | JOB003 |          0.110 |
| DPT003        | JOB004 |          0.190 |
| DPT007        | JOB009 |          0.075 |
+---------------+--------+----------------+
10 rows in set (0.009 sec)
```

No 6 Query
(as Plain Text)

select distinct manager_id
from Employee where manager_id not in (select manager_id
from Employee where salary < 3500);

No 6 SS (of Query & Output in Shell)

```
MariaDB [Company_24141252]> select distinct manager_id
    -> from Employee where manager_id not in (select manager_id
    -> from Employee where salary < 3500);
+------------+
| manager_id |
+------------+
| MNG001     |
| MNG002     |
| MNG003     |
| MNG004     |
| MNG005     |
+------------+
5 rows in set (0.006 sec)
```

No 7 Query (as Plain Text)

select first_name, last_name, employee_id, email, salary, department_id, commission_pct
from Employee e1 where commission_pct = (select min(commission_pct)
from Employee e2 where e2.manager_id = e1.manager_id);

No 7 SS (of Query & Output in Shell)

```
MariaDB [Company_24141252]> select first_name, last_name, employee_id, email, salary, department_id, commission_pct
    -> from Employee e1 where commission_pct = (select min(commission_pct)
    -> from Employee e2 where e2.manager_id = e1.manager_id);
+------------+-----------+-------------+-----------------------+--------+---------------+----------------+
| first_name | last_name | employee_id | email                 | salary | department_id | commission_pct |
+------------+-----------+-------------+-----------------------+--------+---------------+----------------+
| John       | Smith     | EMP001      | john.smith@company.com |  45000 | DPT001        |          0.150 |
| Michael    | Brown     | EMP003      | michael.b@company.com  |  28000 | DPT002        |          0.100 |
| James      | Wilson    | EMP005      | james.w@company.com    |  38000 | DPT003        |          0.180 |
| William    | Garcia    | EMP007      | william.g@company.com  |  32000 | DPT005        |          0.140 |
| Ava        | Walker    | EMP014      | ava.w@company.com      |  31000 | DPT007        |          0.075 |
+------------+-----------+-------------+-----------------------+--------+---------------+----------------+
5 rows in set (0.001 sec)
```